

Lab 7

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6B2

Sample

Client

```
#include <stdio.h>
#include <string.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <unistd.h>
int main()
{
    int sd;
    char msg[100];
    struct sockaddr_in server_addr;
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(4999);
    server_addr.sin_addr.s_addr = inet_addr("127.0.0.1");
    sd = socket(AF_INET, SOCK_STREAM, 0);
    connect(sd, (struct sockaddr *)&server_addr,
    sizeof(server_addr));
    recv(sd, msg, sizeof(msg), 0);
    printf("%s\n", msg);
    close(sd);
    return 0;
}
```

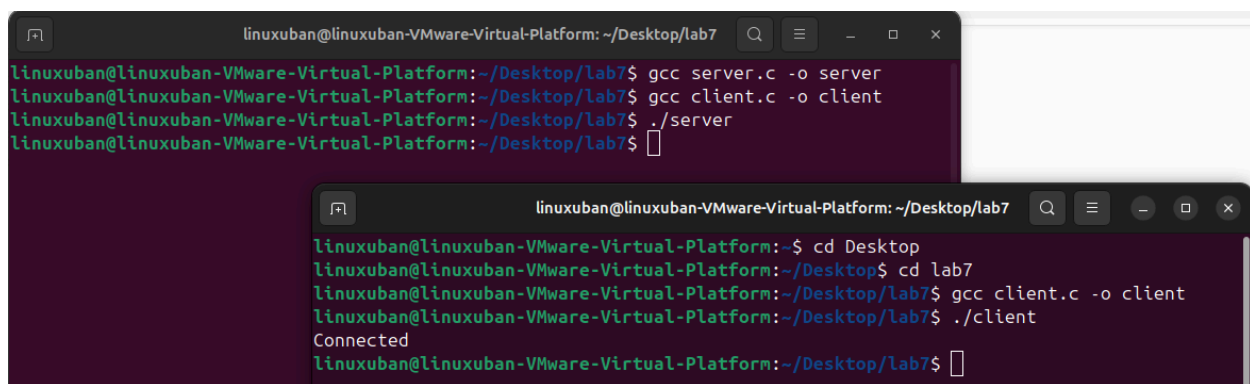
Server

```
#include <stdio.h>
#include <string.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <unistd.h>
int main()
{
    int sd, new_sd;
    char *msg = "Connected";
    struct sockaddr_in my_addr, client_addr;
    socklen_t size;
    my_addr.sin_family = AF_INET;
    my_addr.sin_port = htons(4999);
    my_addr.sin_addr.s_addr = inet_addr("127.0.0.1");
    sd = socket(AF_INET, SOCK_STREAM, 0);
```

```

bind(sd, (struct sockaddr *)&my_addr,
sizeof(my_addr));
listen(sd, 5);
size = sizeof(client_addr);
new_sd = accept(sd, (struct sockaddr *)&client_addr,&size);
send(new_sd, msg, strlen(msg) + 1, 0);
close(new_sd);
close(sd);
return 0;
}

```



The image shows two terminal windows from a Linux virtual machine. The top window shows the compilation of 'server.c' to 'server' and 'client.c' to 'client', followed by running the server. The bottom window shows navigating to the Desktop directory, compiling 'client.c' to 'client', and running the client, which successfully connects to the server.

```

linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab7
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ gcc server.c -o server
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ gcc client.c -o client
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ ./server
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$

linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab7
linuxuban@linuxuban-VMware-Virtual-Platform:~$ cd Desktop
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop$ cd lab7
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ gcc client.c -o client
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ ./client
Connected
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$

```

Q1

Client

```

#include <stdio.h>
#include <string.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <unistd.h>

int main()
{
    int sd;
    int num1, num2, sum;

    struct sockaddr_in server_addr;

```

```

server_addr.sin_family = AF_INET;
server_addr.sin_port = htons(4999);
server_addr.sin_addr.s_addr = inet_addr("127.0.0.1");

sd = socket(AF_INET, SOCK_STREAM, 0);

connect(sd, (struct sockaddr *)&server_addr, sizeof(server_addr));

printf("Enter two numbers: ");
scanf("%d %d", &num1, &num2);

send(sd, &num1, sizeof(num1), 0);
send(sd, &num2, sizeof(num2), 0);

recv(sd, &sum, sizeof(sum), 0);

printf("Server responds: Sum is %d\n", sum);

close(sd);

return 0;
}

```

Server

```

#include <stdio.h>
#include <string.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <unistd.h>

int main()
{
    int sd, new_sd;
    int num1, num2, sum;

    struct sockaddr_in my_addr, client_addr;
    socklen_t size;

    my_addr.sin_family = AF_INET;
    my_addr.sin_port = htons(4999);
    my_addr.sin_addr.s_addr = inet_addr("127.0.0.1");

    sd = socket(AF_INET, SOCK_STREAM, 0);

```

```
bind(sd, (struct sockaddr *)&my_addr, sizeof(my_addr));

listen(sd, 5);

size = sizeof(client_addr);

new_sd = accept(sd, (struct sockaddr *)&client_addr, &size);

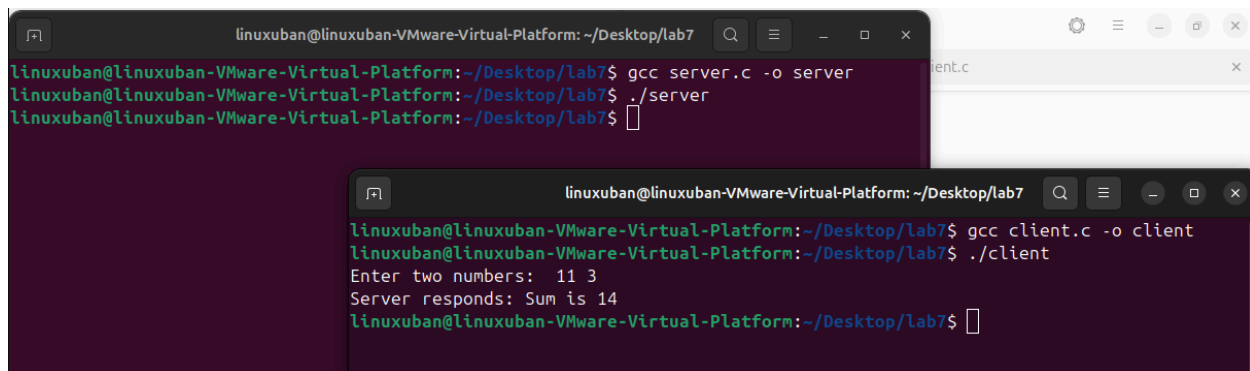
recv(new_sd, &num1, sizeof(num1), 0);
recv(new_sd, &num2, sizeof(num2), 0);

sum = num1 + num2;

send(new_sd, &sum, sizeof(sum), 0);

close(new_sd);
close(sd);

return 0;
}
```



```
linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab7
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ gcc server.c -o server
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ ./server
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$

linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ gcc client.c -o client
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ ./client
Enter two numbers: 11 3
Server responds: Sum is 14
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$
```

Q2

Client

```
#include <stdio.h>
#include <string.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <unistd.h>

int main()
{
    int sd;
    char city[100];
    char msg[100];

    struct sockaddr_in server_addr;

    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(4999);
    server_addr.sin_addr.s_addr = inet_addr("127.0.0.1");

    sd = socket(AF_INET, SOCK_STREAM, 0);

    connect(sd, (struct sockaddr *)&server_addr, sizeof(server_addr));

    printf("Enter city name: ");
    scanf("%s", city);

    send(sd, city, strlen(city)+1, 0);

    recv(sd, msg, sizeof(msg), 0);

    printf("Server Response: %s\n", msg);

    close(sd);

    return 0;
}
```

Server

```
#include <stdio.h>
```

```

#include <string.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <unistd.h>

int main()
{
    int sd, new_sd;
    char city[100];
    char response[100];

    struct sockaddr_in my_addr, client_addr;
    socklen_t size;

    my_addr.sin_family = AF_INET;
    my_addr.sin_port = htons(4999);
    my_addr.sin_addr.s_addr = inet_addr("127.0.0.1");

    sd = socket(AF_INET, SOCK_STREAM, 0);

    bind(sd, (struct sockaddr *)&my_addr, sizeof(my_addr));

    listen(sd, 5);

    size = sizeof(client_addr);

    new_sd = accept(sd, (struct sockaddr *)&client_addr, &size);

    recv(new_sd, city, sizeof(city), 0);

    printf("Client asked weather for: %s\n", city);

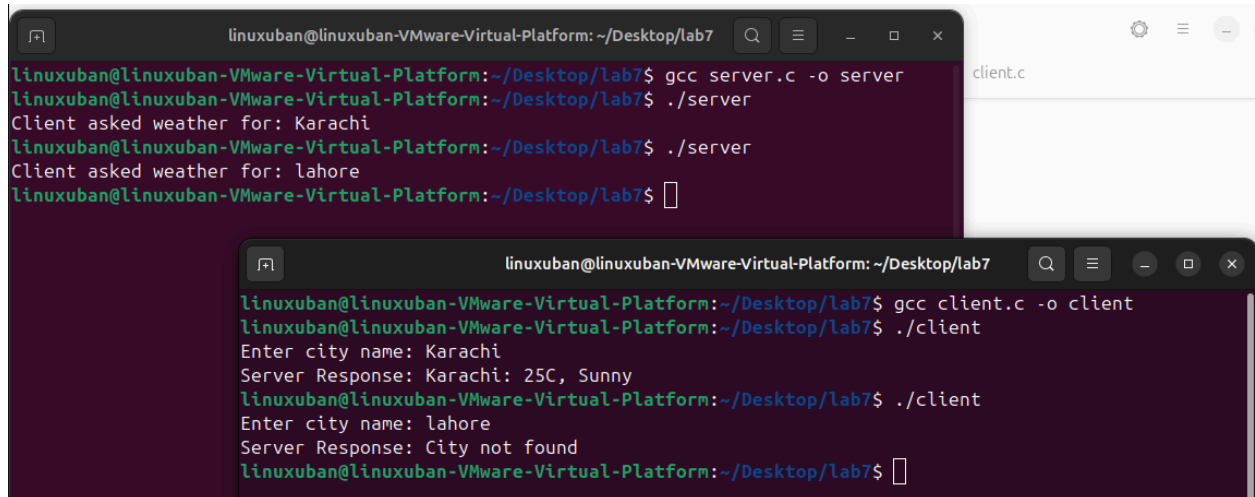
    if(strcmp(city, "Karachi") == 0)
        strcpy(response, "Karachi: 25C, Sunny");
    else if(strcmp(city, "Lahore") == 0)
        strcpy(response, "Lahore: 30C, Hot");
    else if(strcmp(city, "Islamabad") == 0)
        strcpy(response, "Islamabad: 22C, Cloudy");
    else
        strcpy(response, "City not found");

    send(new_sd, response, strlen(response)+1, 0);

    close(new_sd);
}

```

```
close(sd);  
  
return 0;  
}
```



The image shows two terminal windows from a Linux virtual machine. The top window shows the compilation of 'server.c' into 'server' and its execution. The server prints 'Client asked weather for: Karachi' and 'Client asked weather for: lahore'. The bottom window shows the compilation of 'client.c' into 'client' and its execution. The client prompts for a city name, enters 'Karachi' (receiving 'Server Response: Karachi: 25C, Sunny') and 'lahore' (receiving 'Server Response: City not found').

```
linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab7  
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ gcc server.c -o server  
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ ./server  
Client asked weather for: Karachi  
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ ./server  
Client asked weather for: lahore  
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$  
  
linuxuban@linuxuban-VMware-Virtual-Platform: ~/Desktop/lab7  
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ gcc client.c -o client  
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ ./client  
Enter city name: Karachi  
Server Response: Karachi: 25C, Sunny  
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$ ./client  
Enter city name: lahore  
Server Response: City not found  
linuxuban@linuxuban-VMware-Virtual-Platform:~/Desktop/lab7$
```